MTH404 - Dynamics (Quiz 02)

Question # 1 of 10 (Start time: 08:06:19 AM, 09 May 2024)

Which of the following is true about the transverse component of a force in polar coordinates?

Select the correct option

- $F_\theta = m(2\dot{r}\dot{\theta} + 2r\ddot{\theta})$
- $F_\theta = m(2\dot{r}\dot{\theta} - r\ddot{\theta})$
- $F_\theta = m(2\dot{r}\dot{\theta} + r\ddot{\theta})$
- None of thsese

Virtual Bird

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MTH404 - Dynamics (Quiz 02)

Quiz Start Time: 08:06 AM

Question # 6 of 10 (Start time: 08:10:13 AM, 09 May 2024)

Which of the following is true about the radial component of a force in polar coordinates?

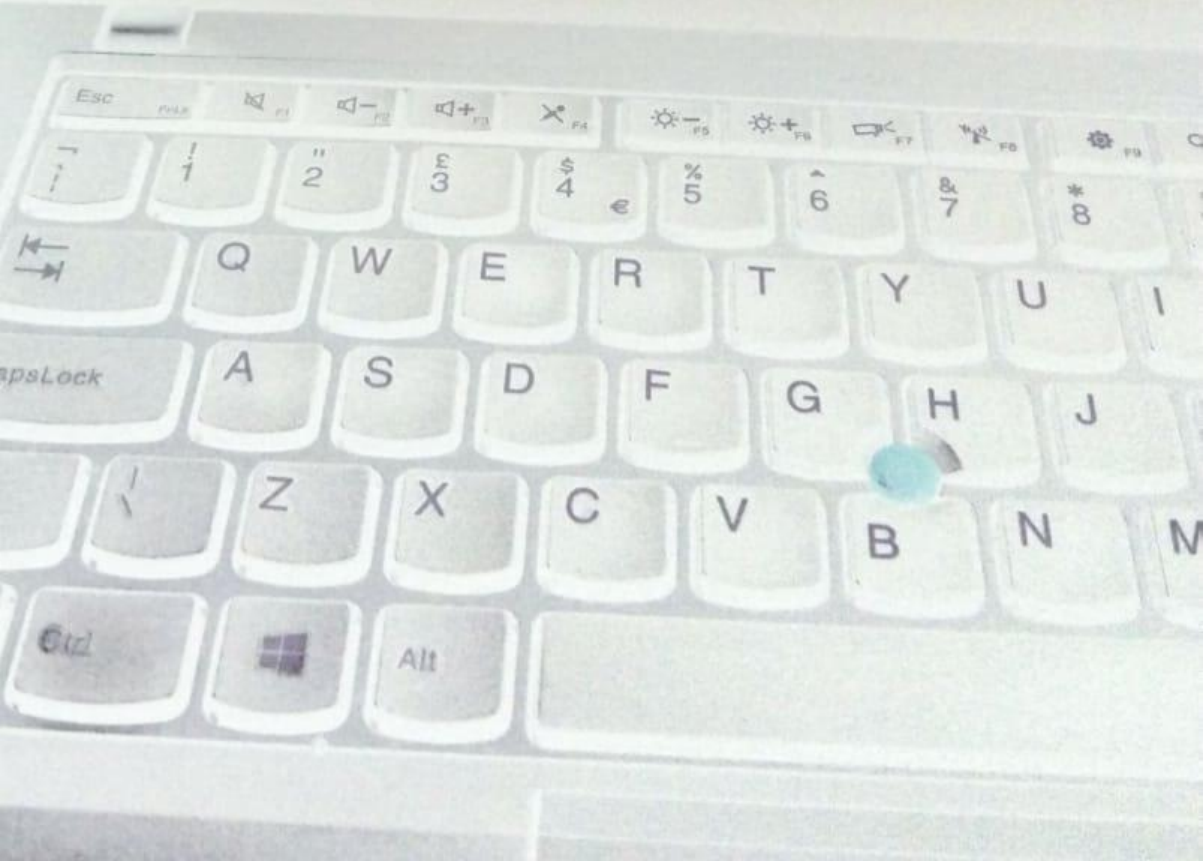
Select the correct option

Reload Math Ec

- $F_r = m(\ddot{r} + r(\dot{\theta})^2)$
- $F_r = m(\ddot{r} - r(\dot{\theta})^2)$
- None of these
- $F_r = m(r(\dot{\theta})^2 - \ddot{r})$

Click to Save Answer & Move to Next Question

Type here to search



Question # 10 of 10 (Start time: 08:15:09 AM, 09 May 2024)

A particle of mass 3kg is moving with velocity 4ti+5j. The force F acting on the particle will be----.

Select the correct option

- 12j
- 12i
- 4i+5j
- 0

Virtulian Bird

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Click to Save Answer & Move to Next Ques

Maps

Question # 5 of 10 (Start time: 08:09:29 AM, 09 May 2024)

At mean position of pendulum, the potential energy of the pendulum is:

Select the correct option

- none of these
- neither maximum nor minimum
- minimum
- maximum

Virtuian Bird

Click to Save Answer & Move to

Type here to search

MTH404 - Dynamics (Quiz 02)

Quiz Start Time: 08:0

Question # 7 of 10 (Start time: 08:11:39 AM, 09 May 2024)

The reciprocal of the time taken by the particle to complete one oscillation is called -----

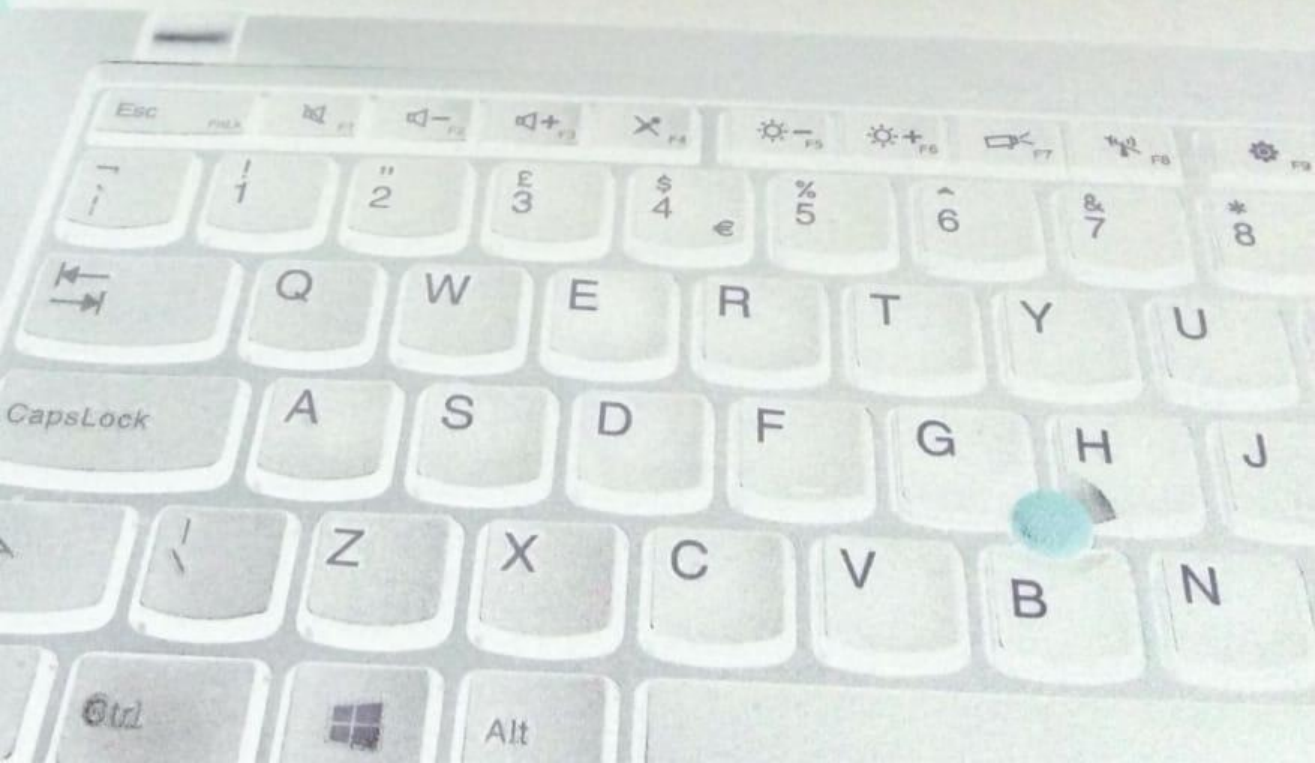
Select the correct option

Reload M

- time period
- frequency
- amplitude
- none of these

Click to Save Answer & Move to Next Qu

Type here to search



MTH404 - Dynamics (Quiz 02)

sec(s)

Quiz Start Time: 08:06 AM, 09 May 202

Question # 8 of 10 (Start time: 08:12:15 AM, 09 May 2024)

Total Marks:

A body of mass 5 g is executing SHM about a point O with amplitude 100 cm. If its max. velocity is 100 cm/sec, its velocity will be 50 cm/sec at a distance (in cm):

Select the correct option

Reload Math Equations

- $50\sqrt{3}$
- $10\sqrt{3}$
- 5
- $50\sqrt{2}$

Click to Save Answer & Move to Next Question

Type here to search

Maps

Question # 4 of 10 (Start time: 08:08:27 AM, 09 May 2024)

frequency is reciprocal of -----.

Select the correct option

- amplitude
- cycle
- wavelength
- time period

Virtual Bird

Click to Save Answer & Move to Next Que

Type here to search

Force acting on a body can change its----.

Select the correct option

- direction of motion
- kinetic energy
- all of the given
- speed

Virtuian Bird

Type here to search

During S.H.M kinetic energy of the body is ----- at mean position.

Select the correct option

- maximum
- none of these
- minimum
- zero

Virtual Bird

Type here to search